

1: Model Used: ChatGPT (GPT-4o)

Prompt:

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letter and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal.

3. Proposed Heuristic: The model proposed adding the current state's rarity score $\text{rarity}(n)$ to the hamming distance baseline $h_0(n)$

$$h_{\text{LLM}}(n) = h_0(n) + \text{rarity}(n)$$

The LLM claimed that because state n is rare, any step originating from n will incur higher path costs, making $h_0(n) + \text{rarity}(n)$ a tighter lower bound for the remaining cost.

2:

If state n is the goal, $h_{\text{LLM}}(n) = 0$

$$h_{\text{LLM}}(n) = h_0(n) + \text{rarity}(n)$$

h_0 = is the number of letters in word n that do not match the goal word

$\text{rarity}(n)$ is the rarity score of the current word n

Pseudocode:

```
def h_llm(node, goal, dictionary):
```

```
    if node.state == goal:
        return 0.0
```

```
    h0 = sum(1 for i in range(len(node.state)) if node.state[i] != goal[i])
    return h0 + dictionary.get(node.state, 0.0)
```

3:

It is not Admissible.

In the frequency metric, cost is incurred when entering a state, not when leaving it. The value $\text{rarity}(n)$ is paid when transitioning into state n . Once inside state n , its rarity score has no effect on future moves. If n is an expensive/rare word, but the remaining path to the goal only visits common words, adding $\text{rarity}(n)$ to the heuristic overestimates the actual remaining cost.

Concrete Counterexample:

Current state (n): vat

Goal state: cat

Dictionary values: Suppose $\text{rarity}(\text{vat}) = 10$ and $\text{rarity}(\text{cat}) = 0$

Hamming distance ($h_0(n)$): 1 (differs by 1 letter)

True Optimal Remaining Cost $h^*(n)$: Transitioning vat $\rightarrow$ cat enters cat costing $1 + \text{rarity}(\text{cat}) = 1.0 = 1$

LLM Heuristic Value: $h_{\text{LLM}}(\text{vat}) = h_0(n) + \text{rarity}(\text{vat}) = 1 + 10 = 11$

Since $h_{\text{LLM}}(\text{vat}) = 11.0 > 1.0 = h^*(\text{vat})$

The heuristic overestimates the optimal remaining cost, proving it is inadmissible.

4:

If word n is the goal: $h_{\text{repaired}}(n) = 0$

Otherwise: $h_{\text{repaired}}(n) = h_0(n) + \text{rarity}(\text{goal})$

Goal state: If you are already at the goal, remaining cost is 0.

Non goal states: Any valid path from state n to the goal state must:

Take at least $h_0(n)$ steps

Finish by entering the goal state, which costs at least $1 + \text{rarity}(\text{goal})$

incur extra non negative costs ≥ 0 for every intermediate word along the path.

Conclusion: The total cost to reach the goal will always be greater than or equal to $h_0(n) + \text{rarity}(\text{goal})$

Because $h_{\text{repaired}}(n)$ never overestimates the actual remaining cost, it is guaranteed to be admissible.